Mid Semester Examination 2025

Name of the Program: M. Tech.(Electric Vehicle Technology) *Semester: First*

Name of the Course: Electrical Machines and Drives *Course Code:* **MEV 104**

Time: 1-1/2 Hour Maximum Marks: 50

Note:

(i) Answer **all the questions** by choosing ***any one of the sub questions***.

(ii) Each question carries 10 marks

Q1	(10 marks)	CO1
(a)	Draw block diagram and describe the essential components of an electrical drive system. Explain the function of each component in detail.	
	OR	
(b)	Discuss the different types of electric drives, i.e, DC drives, AC drives, and special drives. In what situations would one be preferred over the others?	
Q2	(10 marks)	CO2
(a)	What are the merits of electrical drives?	
	OR	
(b)	Discuss the various types of power modulator used in electrical drives.	
Q3	(10 marks)	CO1
(a)	Derive and explain the fundamental torque equation of electrical drives. How does this equation relate motor torque, load torque, and system dynamics?	
	OR	
(b)	A motor drives two loads. One has rotational motion. It is coupled to the motor through a reduction gear with a =0.1 and efficiency of 90%. The load has a moment of inertia of 10 kg-m^2 and torque of 10 N-m. Other load has translatory motion and consists of 1000 kg weight to be lifted up at a uniform speed of 1.5 m/s. Coupling between this load and the motor has an efficiency of 85 %. Motor has an inertial of 0.2 kg-m^2 and runs at a constant speed of 1420 rpm. Determine equivalent inertia referred to the motor shaft and power developed by the motor.	
Q4	(10 marks)	CO2
(a)	Discuss speed–torque conventions and multi-quadrant operation of electric drives. Illustrate your answer with suitable torque-speed characteristics.	
	OR	
(b)	A drive has the following parameters: J=10 kg-m^2, T=100-0.1N, N-m, Passive load torque T_l=0,05N, N-m, where N is the speed in rpm. Initially the drives are operating in steady state. Now it is to be reversed. For this motor characteristic is changed to T=-100-0.1N, N-m. Calculate the time of reversal	CO3
Q5	(10 marks)	CO2
(a)	Explain the concept of load equalization in electrical drives. How does it improve performance during fluctuating load conditions?	
	OR	
(b)	Describe steady-state stability of electrical drives. How can time and energy loss in transient operations be calculated, and why are these parameters important in drive design?	CO3